Sysslan IT Solutions

Internship Cohort

Data Science Task List

About Us

Sysslan IT Solutions

- Sysslan IT Solutions is a leading technology firm recognized for its proficiency in the rapidly evolving domains of cloud computing, blockchain, artificial intelligence (AI), and machine learning (ML).

- The company is dedicated to delivering high-impact projects and customized solutions that address the dynamic requirements of modern businesses.

- Sysslan IT Solutions provides a comprehensive range of products and services, including scalable cloud computing platforms, cutting-edge blockchain solutions, intelligent AI systems, and advanced ML algorithms, designed to empower organizations and drive innovation.

MSME
Micro, Small, & Mediam Enterprises
सूक्ष्म, लघु, एवं मध्यम उद्योग
MSME BUSINESS FORUM

Instructions

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- Update your LinkedIn profile to highlight your accomplishments, such as offer letters or internship certificates from Sysslan IT Solutions. Mention and tag Sysslan IT Solutions in your posts. Use relevant hashtags like #SysslanITSolutions #SysslanExperience #FutureWithSysslan to showcase your professional association.

- Ensure that all work is original and avoid plagiarism or code duplication, as such violations may result in termination of the internship and impact future opportunities with Sysslan IT Solutions.

- Create a video presentation of your completed tasks and share it on LinkedIn. Tag Sysslan IT Solutions and use hashtags like #SysslanInnovation #SysslanITSolutions #SysslanProjects to engage with our professional community.

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SUBMISSION

- Create a professional video highlighting your internship projects and accomplishments.

- Share the video on LinkedIn to demonstrate your work and build credibility with your professional network. Make sure to tag Sysslan IT Solutions in your post so the company is notified of your achievements.

- A submission form will be provided later. In the meantime, continue working on your tasks and maintain separate files for each level of work.

- When posting the video on LinkedIn, use the following hashtags to increase visibility and engagement:
- #SysslanITSolutions #Sysslan #SysslanProjects

- You can also include additional hashtags relevant to your specific internship domain to further showcase your expertise.

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Title: Data Science Project on Train Route Analysis and Journey Time Prediction

Project Description

This project focuses on applying data science techniques to analyze train journey data and build a predictive model for estimating journey duration. The project progresses through data understanding, cleaning, exploration, visualization, and machine learning modeling. By the end of the project, a complete journey time prediction system is developed using Linear Regression, supported by visual and statistical analysis.

Objectives:

- Understand and clean real-world transportation data
- Perform exploratory data analysis and visualization
- Identify patterns and relationships in journey data
- Build and evaluate a predictive machine learning model
- Present insights in a structured and professional manner

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Level 1: Data Overview

Description:

Develop a foundational understanding of the dataset by examining its structure, completeness, and basic statistics.

Tasks:

- Task 1.1: Create a summary of the dataset including total records and columns
- Task 1.2: Prepare a train-wise table showing route start and end stations
- Task 1.3: Calculate basic statistics for distance and number of stops
- Task 1.4: Identify missing, null, or inconsistent values

Skills Gained:

Data understanding, descriptive statistics, Pandas basics, data inspection

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Level 2: Data Cleaning & Feature Engineering

Description:

Clean the dataset and create essential features required for analysis and modeling.

Tasks:

- Task 2.1: Handle missing values and remove duplicate records
- Task 2.2: Standardize arrival and departure time formats
- Task 2.3: Calculate total journey duration for each train
- Task 2.4: Create features such as total distance and number of stops

Skills Gained:

Data cleaning, feature engineering, datetime handling, preprocessing techniques

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Level 3: Data Exploration

Description:
Explore patterns and relationships in the data to gain meaningful insights.

Tasks:

- Task 3.1: Compare journey duration across different routes
- Task 3.2: Identify stations with high and low train traffic
- Task 3.3: Analyze the relationship between distance and journey duration
- Task 3.4: Summarize key exploratory insights

Skills Gained:
Exploratory Data Analysis (EDA), grouping, correlation analysis, analytical thinking

Level 4: Visualization & Pattern Analysis

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Description:
Use visual tools to identify trends, patterns, and relationships within the data.

Tasks:

- Task 4.1: Visualize journey duration across different routes
- Task 4.2: Create charts representing station-wise train traffic
- Task 4.3: Plot distance versus journey duration
- Task 4.4: Explain patterns observed from visual analysis

Skills Gained:
Data visualization, Matplotlib, Seaborn, storytelling with data

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Level 5: Prediction Model Development

Description:

Build and evaluate a machine learning model to predict train journey duration.

Tasks:

- Task 5.1: Prepare a modeling dataset using distance and number of stops as features
- Task 5.2: Split the dataset into training and testing sets
- Task 5.3: Build a Linear Regression model to predict journey duration
- Task 5.4: Evaluate model performance using MAE and RMSE

Skills Gained:

Machine learning basics, Linear Regression, model evaluation, Scikit-learn

Level 6: Final Data Science Project

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Description:

Integrate analysis and modeling into a complete prediction system with visualization.

Tasks:

- Task 6.1: Develop a journey time prediction system using Linear Regression and visualize actual vs. predicted journey duration

Skills Gained:

End-to-end project development, predictive modeling, result visualization, system implementation

How to Contact Us?

To find out more information, please contact us

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